Unit-02: Random variables and probability distributions

1. In a medical examination the chances of error are 15%. Find the Bernoulli distribution if one patient is randomly selected out of 60 patients.

Soln: Let X be the Bernoulli random that represents the number of correct reports.

It is given that 15% of 60 patients have error reports. This means that there are

$\frac{15\%}{60} =$ 9 patients out of 60 who have received error reports. Thus the number of patients receiving

correct reports are 60-9 = 51.

Thus, $P(X = 1) = \frac{51}{60} = 0.85$ and $P(X = 0) = \frac{9}{60} = 0.15$

2. In a quiz contest of answering ‘yes’ or ‘no’, what is the probability of guessing at least 6 answers correctly out of ten question asked? Also find the probability of the same if there are 4 options for a correct answer.

Sol:

$$p = 0.5 \quad q = 0.5 \quad n = 10$$
$$P(x) = {}^{n}C_{x} p^{x} q^{n-x}$$
$$P(x \geq 6) = P(6) + \ldots + P(10) = 0.377$$

When 4 options are given to a correct answer then

$$p = 0.25 \quad q = 0.75 \quad n = 10$$
$$P(x) = {}^{n}C_{x} p^{x} q^{n-x}$$
$$P(x \geq 6) = P(6) + \ldots + P(10) = 0.019$$

3. Assume that the probability of an individual coal miner being killed in a mine accident during a year is 1/2400. Calculate the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year.

Soln: $p = \frac{1}{2400}, \; n = 200, \; m = np = \frac{200}{2400} = \frac{1}{12}$

$P(At\ least\ one\ fatal\ accident\ in\ a\ year) = P(1\ or\ 2\ or\ \ldots\ or\ 200)$

$$= P(1) + P(2) + \cdots + P(200)$$

$$= 1 - P(0) = 1 - \frac{e^{-m}m^0}{0!} = 0.08.$$

4. In a test on 2000 electric bulbs, it was found that the life of a particular make, was normally distributed with an average life of 2040 hours and S.D. of 60 hours. Estimate the number of bulbs likely to burn for
(a) more than 2150 hours, (b) less than 1950 hours and

(c) more than 1920 hours and but less than 2160 hours.

Solution: Here $\mu = 2040\ hours\ \ and\ \ \sigma = 60\ hours$

(a) For $x = 2150, \; z = \frac{x-\mu}{\sigma} = 1.833.$

The number of bulbs to burn more than 2150 hours = 67

Approximately

(b) For $x = 1950, \; z = \frac{x-\mu}{\sigma} = -1.33$

The number of bulbs to burn less than 1950 hours = 184

Approximately

(c) When $x = 1920, \; z = \frac{1920-2040}{60} = -2$

When $x = 2160, \; z = \frac{2160-2040}{60} = 2$

The number of bulbs expected to burn for more than 1920 hours but less than 2160 hours = 1909 nearly.